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Lab – 4 – R Language

Vasukumar_Patel_Lab4.R

Source on Save

```r
taylor <- function(x, numbers){

  approximateValue <- 0

  for (k in 0:(numbers)) {
    approximateValue <- approximateValue + ((-1)^k) * (x-1)^k
  }

  trueValue <- 1 / x
  absoluteError <- abs(trueValue - approximateValue)

  relativeError <- absoluteError / abs(trueValue)

  cat("x value:", x, "\n")
  cat("n value:", numbers, "\n\n")

  cat("True value:        ", trueValue, "\n")
  cat("Approximate value: ", approximateValue, "\n")
  cat("Absolute error:    ", absoluteError, "\n")
  cat("Relative error:    ", relativeError, "\n\n")

}

taylor(0.1,10)
taylor(0.1,100)

taylor(1.75,10)
taylor(1.75,100)

taylor(2.1,10)
taylor(2.1,100)

taylor(3.3,10)
taylor(3.3,100)
```

Console | Terminal | Background Jobs

R 4.5.2 · ~/

```
> taylor(0.1,10)
x value: 0.1
n value: 10

True value:         10
Approximate value:  6.861894
Absolute error:     3.138106
Relative error:     0.3138106

> taylor(0.1,100)
x value: 0.1
n value: 100

True value:         10
Approximate value:  9.999761
Absolute error:     0.0002390526
Relative error:     2.390526e-05

>
> taylor(1.75,10)
x value: 1.75
n value: 10

True value:         0.5714286
Approximate value:  0.5955629
Absolute error:     0.02413436
Relative error:     0.04223514

> taylor(1.75,100)
x value: 1.75
n value: 100

True value:         0.5714286
Approximate value:  0.5714286
Absolute error:     1.373346e-13
Relative error:     2.403355e-13

>
```

```
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>
> taylor(2.1,10)
x value: 2.1
n value: 10

True value:          0.4761905
Approximate value:   1.834817
Absolute error:      1.358627
Relative error:      2.853117

> taylor(2.1,100)
x value: 2.1
n value: 100

True value:          0.4761905
Approximate value:   7218.892
Absolute error:      7218.416
Relative error:      15158.67

>
> taylor(3.3,10)
x value: 3.3
n value: 10

True value:          0.3030303
Approximate value:   2887.605
Absolute error:      2887.302
Relative error:      9528.098

> taylor(3.3,100)
x value: 3.3
n value: 100

True value:          0.3030303
Approximate value:   1.037522e+36
Absolute error:      1.037522e+36
Relative error:      3.423824e+36
```

1. What is happening to the accuracy of the approximation as n increases? Explain with examples your reasoning.

➔ As n increases, the accuracy depends on the value of x.

➔ When $0 < x < 2$, the approximation becomes more accurate.
This is because the terms get smaller and the error decreases.
Example: for x = 0.1 and x = 1.75, the result is much better when n = 100 than n = 10.

➔ When $x > 2$, the approximation becomes less accurate.
This is because the terms get bigger and the error increases.
Example: for x = 2.1 and x = 3.3, the result becomes very wrong when n increases.

2. What is happening to the accuracy of the approximation as x gets further away from the value of a (which is 1 for this lab)? Explain your reasoning with examples.

➔ As x gets further away from a = 1, the approximation becomes less accurate.

➔ Examples:

➔ x = 1.75
→ The approximation is very accurate, especially when n is large.

➔ x = 0.1
→ The approximation is less accurate at first, and needs a larger n to improve.

➔ x = 2.1 and x = 3.3
→ The approximation becomes very bad, even with large n.

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3. Which would be considered better for increasing the accuracy of the approximation? A larger value for n? Or values of x closer to a? Explain your decision.

➔ Values of x closer are better for increasing accuracy.

➔ Even if you make n larger, the approximation only improves when x is close to 1. If x is far from 1, increasing n may not help and can even make the result worse.

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4. Explain what is happening at x = 3.3 for n = 100.

➔ At x = 3.3 and n = 100, the approximation becomes very large and completely inaccurate because the values being added in the series keep getting bigger instead of smaller. This happens because x is far from 1, so the method no longer works properly. Instead of getting closer to the true value (which is about 0.303), the result keeps moving further away as n increases. This means the series is not converging and the approximation becomes worse when n is large.

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5. Explain why the sign is changing for the approximate value at x = 3.3 for n = 99.

➔ At x = 3.3, the sign changes because the formula includes (-1)^k, which makes each term alternate between positive and negative. So one term is added, the next is subtracted, and this pattern continues. When n = 99, the last term has an odd power, so it becomes negative, which can make the final approximate value negative. Since the terms are very large at this x value, the last term has a strong effect on the result, causing the overall sign to flip.